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--- Full Doppler Velocity Pipeline with N-frame Window Averaging ---

```
clear
load('matlab_motion_.5v_5_1.mat'); % 1024x1024x256
```

--- Parameters ---

```
N = 10; % number of frames per averaging window
T = 0.2; % time between successive A-scans (s)
f0 = 455e-9; % system frequency (Hz)
theta = 0; % angle in radians for velocity calculation
n = 5; % parameter for velocity scaling
```

```
[rows, cols, frames] = size(im_data);
```

```
N =
```

```
10
```

--- Step 1: Hilbert transform ---

```
dop = hilbert(im_data);
I = real(dop);
Q = imag(dop);
```

--- Step 2: Compute average phase shift over N-frame windows ---

```
avgPhaseShift = []; % initialize
```

```
k = 1; while k + N - 1 <= frames numerator = zeros(rows, cols); denominator = zeros(rows, cols);
```

```

    % Sum over N-1 consecutive pairs within this window
    for i = 0:(N-2)
        numerator = numerator + ( Q(:, :, k+i) .* I(:, :, k+i+1) - I(:, :, k+i) .* Q(:, :, k+i+1) )
        denominator = denominator + ( I(:, :, k+i) .* I(:, :, k+i+1) + Q(:, :, k+i) .* Q(:, :, k+i+1) )
    end

    avgPhaseShift(:, :, end+1) = -atan(numerator ./ denominator); % append to array
    k = k + N; % jump by N frames for the next window
end

avgPhaseShift = []; % initialize

for k = 1:(frames - N + 1) % sliding window from frame 1 to frame (frames-
N+1)
    numerator = zeros(rows, cols);
    denominator = zeros(rows, cols);

    % Sum over N-1 consecutive pairs within this window
    for i = 0:(N-2)
        numerator = numerator + ( Q(:, :, k+i) .* I(:, :, k+i+1) - I(:, :, k+i) .*
Q(:, :, k+i+1) );
        denominator = denominator + ( I(:, :, k+i) .* I(:, :, k+i+1) +
Q(:, :, k+i) .* Q(:, :, k+i+1) );
    end

    avgPhaseShift(:, :, end+1) = -atan(numerator ./ denominator); % append to
array
end

```

--- Step 3: Doppler frequency ---

```
fd = avgPhaseShift / (2*pi*T);
```

--- Step 4: Velocity ---

```
velocity = (f0 ./ (2*1.33*cos(theta))) .* fd;
```

--- Step 5: Average velocity per frame/window ---

```
D_doppler = squeeze(mean(mean(velocity, 1), 2)) * 10e9; % 1D array, one value
per N-frame window
```

--- Step 6: Generate PZT reference ---

--- Step 7: Time vector ---

```
time_t = (1:length(D_doppler)) * T * N; % account for N-frame windowing
```

--- Step 8: Plot Doppler vs PZT ---

```
figure;
d_const = ones(1, length(D_doppler))
plot(time_t, D_doppler, 'b', 'LineWidth', 1.5);
plot(time_t(1:length(d_const)),circshift(0.5*1000*(d_const)*2.8/75,6));
xlabel('time (s)');
ylabel('velocity (nm/s)');
legend('doppler estimated', 'pzt');
title('Average Doppler Velocity vs PZT Reference');
grid on;
```

d_const =

Columns 1 through 13

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 14 through 26

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 27 through 39

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 40 through 52

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 53 through 65

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 66 through 78

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 79 through 91

1 1 1 1 1 1 1 1 1 1 1 1
1

Columns 92 through 104

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 105 through 117

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 118 through 130

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 131 through 143

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 144 through 156

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 157 through 169

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 170 through 182

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 183 through 195

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 196 through 208

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 209 through 221

1 1 1 1 1 1 1 1 1 1 1 1

1

Columns 222 through 234

1 1 1 1 1 1 1 1 1 1 1 1

1

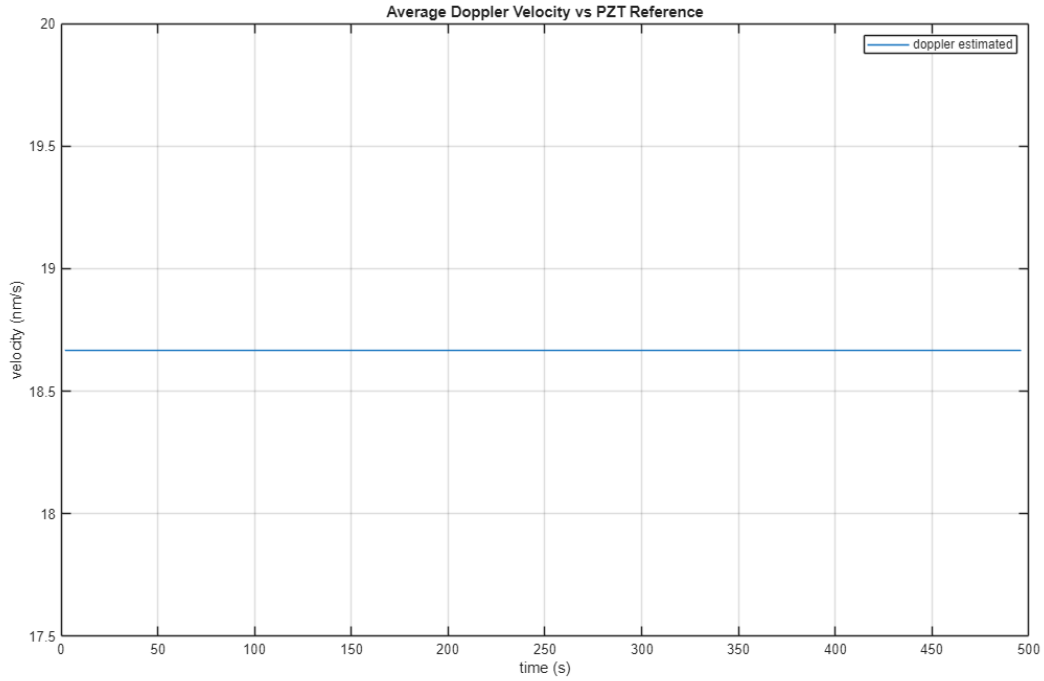
Columns 235 through 247

1 1 1 1 1 1 1 1 1 1 1 1

Column 248

1

Warning: Ignoring extra legend entries.



```
N_size=1024; x=zeros(N_size);x0=1:1:N_size; for i=1:1:N_size x(:,i)=x0; end y=x'; %% data_input=im_data;
N_window=256; data1d=abs(fft(permute(data_input(512,512,1:N_window),[3 1 2])));
N0=4;[max_fft,peak_index]=max(data1d(N0:N_window/2));
data3d_fd=ifft(data_input(:,1+N_window*0:N_window*1),N_window,3);
%% roi_top=128;roi_bottom=1024-128; roi_left=128;roi_right=1024-128;
%% N_interval=100;% interval between data points used to calculate Doppler phase shift
data3d_fd_half=data3d_fd;
data3d_fd_half(:,1:2)=0;data3d_fd_half(:,N_window/2+1:N_window)=0;
data3d_fd_half=circshift(data3d_fd_half,[0 0 -(peak_index+N0-1-1)]);
data3d_fd_half=fft(data3d_fd_half,[],3);
data_doppler_3D=angle(data3d_fd_half(:,1+N_interval:N_window).*conj(data3d_fd_half(:,1:N_window-N_in-
terval)));
data_doppler_3D(data_doppler_3D>0.9*pi)=data_doppler_3D(data_doppler_3D>0.9*pi)-pi;
```

```

data_doppler_3D(data_doppler_3D<-0.9*pi)=data_doppler_3D(data_doppler_3D<-0.9*pi)+pi;

figure for i=1:1:size(N_window-N_interval,1) imagesc(data_doppler_3D(:,i)) drawnow end

N=19; d_doppler=permute(data_doppler_3D(330,527,:),[3 1 2])*455/1.33/4/pi/N_interval/0.2; for
i=1:1:floor(size(d_doppler, 1)/N/2) for j=1:1:N d_const((i-1)*N*2+j)=1; end

    for j=1:1:N
        d_const((i-1)*N*2+j+N)=-1;
    end
end
%%

time_t=(1:1:size(d_doppler, 1))*0.2; figure plot(time_t,d_doppler) hold on plot(time_t(1:length(d_const)),circ-
shift(0.5*1000*(d_const)*2.8/75,6)) xlabel('time') ylabel('velocity(nm/s)') legend ('doppler estimated','pzt')

```

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